AI: THE GLOBAL TRANSFORMATION

Project Planning & Management | Literature Review | Requirements Gathering

DEPI – Data Analysis Track | Round 5
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Submission Date: 16 October 2026

Prepared as a data-analysis project proposal. All future findings are provisional until verified through collected data and documented analysis.

Document Guide

This combined document contains the three DEPI deliverables, written in academic English.

- Part I — Project Planning & Management
- Part II — Literature Review
- Part III — Requirements Gathering

Project Snapshot

Item	Description
Main headline	AI: The Global Transformation
Descriptive title	The AI Transformation: Investment, Jobs, Economic Growth, Business Competition, and Human Well-being
Project type	Exploratory data analysis and interactive data visualization
Primary focus	AI investment over time; relationship between AI and economic indicators; AI adoption and business performance
Proposed tools	Excel for data inspection; SQL for structured data; Python for cleaning and exploratory/statistical analysis; Power BI for interactive visualization; GitHub for version control and documentation
Intended output	Documented data workflow, reproducible analysis, interactive dashboard, report of findings and limitations
Forecast horizon	Up to 2030 only where a suitable source provides defensible historical data and forecasting is appropriate; projections will be labelled separately from observed data

PART I — PROJECT PLANNING & MANAGEMENT

1.1 Project Overview

Artificial intelligence is attracting investment and being adopted across industries, while its potential implications for productivity, economic development, employment, and competition remain uneven across countries and firms. This project will use publicly available, documented data to explore how AI investment changes over time, how AI-related indicators relate to selected economic measures, and what patterns can be observed between AI adoption and business performance.

The project is analytical rather than causal. Unless a dataset or research design supports causal inference, observed relationships will be described as associations and will not be presented as proof that AI alone caused a company to succeed or fail.

1.2 Problem Statement

AI investment and adoption are expanding, but headline claims about their economic and business effects can be difficult to compare because datasets use different definitions, time periods, geographic coverage, and measurement methods. The project addresses the need for a transparent, data-driven view of selected AI investment, economic, and company-performance indicators, while making data limitations and uncertainty explicit.

1.3 Aim and Objectives

Overall aim: To conduct an evidence-based exploratory analysis of AI investment trends and their relationships with selected economic and business-performance indicators.

- Measure and visualize AI investment trends across available years, regions, and sectors where comparable data exist.
- Compare selected AI-related indicators with documented economic measures such as productivity, output, or employment indicators, without assuming causation.
- Explore business-level patterns using available indicators such as AI adoption, revenue growth, productivity, market share, or operating performance, subject to data availability.
- Build a documented data pipeline and interactive Power BI dashboard that clearly identifies sources, definitions, filters, and limitations.
- Translate the analysis into cautious, evidence-based insights and recommendations for further investigation.

1.4 Research Questions

1. How have AI-related investment levels changed over time, and how do observed trends differ across regions or sectors where comparable data are available?
2. What relationships, if any, appear between AI-related indicators and selected measures of economic activity, productivity, or employment?
3. What patterns are observable between AI adoption and company-performance indicators, and how do these patterns vary across firms or sectors?
4. What data limitations prevent the project from concluding that AI directly caused a particular economic outcome or company success/failure?

1.5 Scope and Boundaries

Included	Excluded / Limitations
Public, reputable, and documented datasets; analysis of historical trends; descriptive statistics; comparisons; visualizations; transparent interpretation.	No unsupported claims that AI caused a company to collapse or succeed; no private or unlawfully obtained data; no guaranteed predictions.
Geographic and sector comparisons only where definitions and coverage are sufficiently comparable.	No forced comparison between indicators with incompatible definitions, currencies, periods, or units.
Forecasting to 2030 only if appropriate data and a defensible method are available; forecast assumptions will be stated.	Forecasts will not be presented as facts or as guaranteed outcomes.
Company examples may be used if reliable public company data and AI-adoption evidence can be linked with appropriate caveats.	Anecdotal stories alone will not be treated as statistical evidence.

1.6 Proposed Methodology and Tools

Stage	Activities	Proposed tools / outputs
1. Source discovery	Identify public datasets and research sources; record publisher, date, definitions, geographic coverage, and license/access conditions.	Source register; Excel/CSV
2. Data inspection	Check structure, missingness, duplicates, units, date ranges, outliers, and consistency.	Excel; Python
3. Data preparation	Clean fields, standardize dates and units where justified, document transformations, and preserve raw copies.	Python; SQL
4. Exploratory analysis	Calculate descriptive statistics, year-on-year changes where valid, group comparisons, and correlations when appropriate.	Python; SQL
5. Visualization	Build trend lines, comparison charts, scatter plots, and KPI cards with clear labels and source notes.	Power BI
6. Interpretation and QA	Check whether visuals match underlying data; review assumptions; distinguish association from causation.	Peer review; validation checklist
7. Delivery	Submit report, dashboard, data dictionary, source log, and code/notebooks where used.	Report; Power BI file; GitHub repository if approved

1.7 Project Plan and Milestones

The schedule below follows the milestone dates provided in the DEPI brief. The team should confirm whether each date is a submission deadline or a target review date.

Milestone / deliverable	Target date	Main tasks	Completion evidence
Project Planning & Management	16 Oct 2026	Confirm scope, aim, questions, roles, timeline, risks, KPIs, and tools.	Approved planning section
Literature Review	16 Oct 2026	Review credible reports and studies; summarize findings and limitations; identify the project gap.	Cited literature review and reference list
Requirements Gathering	16 Oct 2026	Identify stakeholders, user stories, use cases, functional and non-functional requirements.	Requirements specification
System Analysis & Design	6 Nov 2026	Design data model, data flow, dashboard layout, metrics, and validation approach.	Design diagrams/mock-ups and data dictionary
Implementation (Source Code & Execution)	30 Nov 2026	Collect and clean data; perform analysis; build visuals; test calculations and filters.	Working analysis, code and dashboard
Final Presentation, Testing & Reports	4 Dec 2026	Run final QA, document results and limitations, prepare slides and demonstrate dashboard.	Final report, tested dashboard and presentation

1.8 Gantt-Style Work Schedule

Indicative schedule based on the supplied deadlines; detailed weekly allocation can be adjusted after the team confirms dataset availability.

Workstream	16 Oct	17–25 Oct	26 Oct–6 Nov	7–20 Nov	21–30 Nov	1–4 Dec
Planning and scope	●	Review				
Literature and source register	●	●	Update			Final citations
Requirements gathering	●	Review	Confirm			
Data discovery and collection	Start	●	●	Complete		Refresh if needed
Data cleaning and analysis		Plan	Design	●	Complete	Validate
Dashboard and report		Outline	Mock-up	Build	Test	Finalize
Presentation and final QA				Plan	Draft	●

1.9 Team Roles and Task Assignment

The team consists of five members, with Omima Mohamed Alam El-Din as team lead, as recorded in the original project file. The role allocation below is a proposed starting point and should be agreed by the team; it does not claim that individual tasks have already been completed.

Team member	Proposed responsibility	Expected contribution
Omima Mohamed Alam El-Din — Team Lead	Coordination and integration	Track deadlines, coordinate meetings, maintain scope, integrate sections, conduct final consistency review.
Sayed Ahmed Sayed	Data sourcing and documentation	Find candidate datasets, document source metadata, definitions, and data availability.
Eman Abdullah	Data cleaning and quality checks	Support missing-value, duplicate, consistency, and transformation checks.

Rana Raafat Abdelhamid	Analysis and literature support	Summarize selected studies and support descriptive/statistical analysis.
Osama Ayman Hosny	Visualization and testing support	Support dashboard layout, chart validation, usability checks, and presentation assets.

All members should review the final conclusions and sources. Roles are flexible and may be reassigned based on availability and each member's skills.

1.10 Risk Assessment and Mitigation

Risk	Likelihood / impact	Mitigation	Owner / review
Relevant data are unavailable, incomplete, or not comparable.	Medium / High	Begin source discovery early; define minimum viable indicators; record gaps; narrow scope instead of filling gaps with assumptions.	Data sourcing lead; weekly
AI adoption and company performance cannot be linked at firm level.	High / High	Treat business-level analysis as conditional on data availability; use sector-level analysis or documented case studies if necessary; avoid causal claims.	Team lead + analysis lead
Conflicting definitions across sources.	Medium / High	Maintain a data dictionary; record units, scope, and methodology; do not combine incompatible measures.	Data quality lead
Deadline pressure and uneven task distribution.	Medium / High	Use milestone check-ins, shared task tracker, small review deadlines, and backup ownership for critical tasks.	Team lead; twice weekly near deadlines
Incorrect calculations or misleading charts.	Medium / High	Validate totals and transformations, cross-check selected calculations, label axes/units, and perform peer review.	Analysis + visualization leads
Overstating causality or making unsupported predictions.	Medium / High	Use association language; state limitations; separate observed data from forecasts and interpretations.	All members; final review
Data licensing, privacy, or reproducibility concerns.	Low–Medium / High	Use permitted public sources; record license/terms; do not expose personal or confidential data; retain a source log and transformation notes.	Team lead

1.11 Key Performance Indicators (KPIs)

KPI	Proposed success criterion	How it will be checked
Source traceability	Every analytical indicator has a recorded source, definition, unit, and retrieval date.	Audit source register and dashboard notes.
Data quality documentation	Missing values, duplicates, and key cleaning decisions are documented for each dataset.	Data quality checklist and transformation log.
Calculation validation	All headline figures and displayed calculations are independently checked before submission.	Recalculate sample totals/ratios and compare with dashboard outputs.
Reproducibility	Another team member can rerun the documented preparation steps using the supplied files and instructions, where tooling permits.	Peer rerun of a sample workflow.
Dashboard usability	Core questions can be answered using clearly labelled visuals and filters, without unexplained metrics.	Team walkthrough and feedback checklist.
Deadline completion	All three initial deliverables are prepared for the 16 Oct deadline; later milestones are tracked against the supplied schedule.	Milestone tracker.
Evidence-based conclusions	Every major conclusion is linked to a chart, table, or cited source and includes limitations where needed.	Final report review.

These are proposed project-quality KPIs, not results already achieved. The team can set numeric thresholds after confirming the datasets and dashboard scope.

PART II — LITERATURE REVIEW

2.1 Purpose and Review Approach

This review establishes an evidence base for three connected themes: AI investment, economic development, and business performance. It uses a small initial set of identifiable, credible sources to frame the project. It is a preliminary literature review, not a claim to be exhaustive. The team should expand it after selecting the final datasets and keywords.

2.2 AI Investment Trends

The Stanford Institute for Human-Centered Artificial Intelligence (HAI) AI Index Report 2025 documents a substantial expansion in AI-related investment. It reports that global corporate AI investment reached USD 252.3 billion in 2024, with private investment increasing 44.5% compared with 2023. It also reports that private investment in generative AI reached USD 33.9 billion in 2024. These figures indicate that investment is a measurable dimension of AI's expansion, but they should be interpreted according to the report's definitions and coverage rather than treated as a complete measure of all AI spending worldwide (Stanford HAI, 2025).

For this project, the report provides a credible starting point for identifying investment indicators and possible year-to-year comparisons. The analysis should distinguish private investment, corporate investment, mergers and acquisitions, public funding, and other categories wherever the source separates them. Combining these categories without checking definitions could create misleading trends.

2.3 AI and Economic Development

The International Monetary Fund's staff discussion note on generative AI and the future of work discusses the potential for AI to reshape labor markets and economic outcomes. It highlights that exposure to AI differs across economies and workers, and that productivity gains, distributional effects, and inequality are important considerations. This suggests that AI's economic implications should not be reduced to a single growth figure: the distribution of benefits and the structure of employment also matter (Cazzaniga et al., 2024).

The project can translate this broad discussion into measurable questions by comparing selected AI-related indicators with documented economic measures. Candidate measures may include labor productivity, GDP growth, employment indicators, or sector-level output, but the final selection must depend on source coverage, comparable definitions, and sufficient observations. A correlation or trend comparison can identify patterns; by itself, it cannot prove that AI caused economic growth or employment changes.

2.4 AI Adoption and Business Performance

Brynjolfsson, Li, and Raymond's study, "Generative AI at Work," examines the introduction of a generative AI assistant among 5,179 customer-support agents. The study reports an average increase of approximately 14% in issues resolved per hour, with larger gains for less experienced and lower-skilled workers and limited effects for some more experienced workers. The study is useful because it measures a specific workplace setting rather than relying only on broad claims about AI (Brynjolfsson et al., 2025; earlier NBER working paper issued in 2023).

However, this is evidence about worker productivity in a particular customer-support context, not proof that every AI-adopting company will grow or that non-adopting companies will fail. Company performance depends on multiple factors, and public data may not provide a consistent measure of AI adoption and business outcomes across firms. Therefore, the project should look for observable patterns and clearly describe the limits of any company comparisons.

2.5 Synthesis of the Initial Literature

Theme	What the selected sources support	Implication for this project
Investment	AI investment has expanded substantially in recent years; measurement depends on investment category and source definitions.	Use a year-based dataset with clearly defined investment categories and documented coverage.
Economy and labor	AI may affect productivity and labor-market outcomes, with different effects across workers and economies.	Compare carefully selected indicators; avoid assuming that all groups benefit equally.
Business performance	A specific workplace study found productivity gains from an AI assistant, but effects varied by worker group and context.	Use company/sector data cautiously and avoid generalizing a single case to all businesses.

2.6 Research Gap and Project Contribution

The initial sources provide important but different types of evidence: investment statistics, macro-level discussion of labor-market implications, and a specific study of AI-assisted work. The project's proposed contribution is to bring selected investment, economic, and business indicators into a transparent exploratory analysis and dashboard, while documenting how measures are defined and where comparisons are limited. This is a practical analytical contribution, not a claim that no previous research has studied these relationships.

2.7 Preliminary Hypotheses / Analytical Expectations

The following are questions to test, not established findings:

- H1: AI-related investment indicators show an upward trend over the selected period, subject to the chosen dataset and investment definition.
- H2: Selected AI-related indicators may be associated with selected economic indicators, but the direction and strength must be tested on the data.
- H3: Companies or sectors with documented AI adoption may show different performance patterns, but differences may reflect industry, firm size, strategy, market conditions, or other confounding factors.

2.8 Limitations of the Review

This initial review uses a limited number of sources and does not yet cover every theme in the original project concept, such as public opinion, social well-being, skills, and the full range of business models. The final review should be updated once the team confirms the exact variables and datasets. Any source-specific statistics should be checked against the source report before final presentation.

REQUIREMENTS GATHERING

AI: The Global Transformation

AI Impact Intelligence Dashboard

1) Stakeholder Analysis

A stakeholder is a person or organization that influences the project, is affected by it, supplies data or constraints, evaluates it, or uses its outputs. Analysis is needed because a visually attractive dashboard can still fail if it does not answer the right questions.

Some important Topic:

- Stakeholder
- Role / interest
- Needs
- Engagement / evidence

2) User Stories & Use Cases

2.1- User Story:

A User Story states a need from the user's point of view: “As a [user], I want [capability], so that [benefit].” It needs acceptance criteria: observable conditions that show the story works.

2.1- User Case:

A Use Case describes a complete interaction to achieve a goal. It identifies the actor, trigger, preconditions, main steps, exceptions, postconditions, and evidence. Stories state the need; use cases explain the sequence and failure paths.

3) Functional Requirements

Functional requirements describe system capabilities. Each should have a unique ID, be feasible and testable, and connect to a user need/use case. Items below are proposed initial-release requirements; confirm them after data inventory and supervisor review.

4) Non-functional Requirements

Non-functional requirements define quality and operating constraints. Thresholds below are proposed student-project targets and must be checked against actual dataset size, hardware, Power BI environment, and supervisor expectations.

1) Stakeholder Analysis:

Stakeholder	Role / interest	Needs	Engagement / evidence
Supervisor / DEPI evaluator	Approves direction and evaluates deliverables.	Feasible scope, alignment with DEPI expectations, clear evidence and documentation.	Review proposal and requirements; record feedback, decisions, and approval.
Project team (five members)	Builds, tests, documents, presents.	Clear ownership, shared definitions, manageable tasks, reproducible files.	Weekly check-in, task board/GitHub issues, peer review.
Analyst / researcher	Primary analytical user.	Reliable source metadata, filters, definitions, reproducible transformations, underlying values.	Review analysis questions and prototype; test workflows.
General audience / presentation viewer (Those who are concerned about the future and fearful of artificial intelligence)	Wants a comprehensible overview.	Plain-language explanation, visible sources, clear chart titles and uncertainty.	Walkthrough with non-technical reviewer.

2.1- User Stories:

ID	User story	Acceptance criteria / example
US-01	As an analyst, I want overview KPIs so I can understand the main indicators quickly.	Each KPI has label, unit, period, and source context. Unavailable data is labelled unavailable, not invented.
US-02	As a researcher, I want to filter by year, country, and sector where available so I can compare like with like.	Filters update relevant visuals; selected filters are visible; missing categories are not silently misclassified.
US-03	As a reviewer, I want to trace a chart to its source so I can verify it.	Each published chart has attribution; key metrics link to source metadata and transformation notes.
US-04	As an analyst, I want annual investment comparisons so I can identify changes over time.	Definition, unit/currency, period, and nominal/adjusted status are shown; missing years are not shown as zero without source support.
US-05	As a user, I want to inspect survey findings on use or trust.	Show source, field dates, sample/geography if supplied, and question wording or a link.
US-06	As a non-technical viewer, I want clear chart titles and labels.	Charts state what is measured, unit, period; acronyms are defined in a glossary or tooltip.

2.2- Use Cases:

UC-01 — Explore the overview dashboard

Field	Description
Actor / goal	Analyst, supervisor, or general viewer; understand key indicators and navigate to a theme.
Preconditions / trigger	Dashboard is available and the approved dataset has been loaded; user opens the dashboard.
Main flow	1. Show overview KPIs. 2. Display label, unit, period. 3. User selects a theme/visual. 4. Show relevant detail. 5. Make source notes available.
Exception	If refresh fails or a dataset is missing, show clear status or omit affected KPI with explanation; never show fabricated/current-looking data.
Postcondition / evidence	User can reach theme details. Demonstrate navigation and verify one KPI against its calculation.

UC-02 — Filter and compare indicators

Field	Description
Actor / goal	Analyst/researcher; compare a measure across years and available country/sector/company dimensions.
Preconditions / trigger	Selected data contains the needed dimensions; user changes a filter or selects a chart element.
Main flow	1. Choose metric. 2. Select time range. 3. Choose available dimension. 4. Review updated chart. 5. Reset or compare another group.
Exception	If no records match, show an empty-state message, not zero. Hide dimensions absent from the data.
Postcondition / evidence	The filtered comparison has clear scope. Test a known filter combination against SQL/source values.

UC-03 — Trace a chart to its source

Field	Description
Actor / goal	Supervisor/researcher; verify the origin and calculation of an indicator.
Preconditions / trigger	Source metadata and transformation notes exist; user opens source details.
Main flow	1. Identify metric. 2. Open source details. 3. Read publisher, title, URL, coverage, unit, access date. 4. Review transformation. 5. Reproduce a sample calculation where feasible.
Exception	If URL is broken, retain citation metadata and log the broken link; replace the link if possible.
Postcondition / evidence	User can understand and, where feasible, reproduce the metric. Trace one KPI end-to-end.

UC-04 — Review a forecast

Field	Description
Actor / goal	Analyst/supervisor; assess a selected forecast and its limitations.
Preconditions / trigger	Historical data supports a model and a holdout evaluation; user opens forecast page.
Main flow	1. View observations. 2. View forecast separately. 3. Review horizon, model, evaluation metric. 4. Read assumptions. 5. Compare with baseline if available.
Exception	If series is too short/inconsistent or model fails evaluation, do not publish a confident forecast; mark unavailable or show a justified baseline.
Postcondition / evidence	User distinguishes observations from estimates. Verify time-aware train/test separation and chart labels.

UC-05 — Refresh and validate data

Field	Description
Actor / goal	Team analyst; update the dataset safely.
Preconditions / trigger	Permitted source files/access and processing instructions are available; new approved data version arrives.
Main flow	1. Preserve raw source where allowed. 2. Run cleaning. 3. Execute quality checks. 4. Load validated data. 5. Compare totals. 6. Record version/results.
Exception	If schema or quality checks fail, stop publication, log issue, and retain last validated output with its date clearly shown.
Postcondition / evidence	Data is either validated and released or blocked with a documented issue. Inspect logs and quality report.

1) Functional Requirements:

Implement all of these features if possible:

ID	Feature
FR-01	Source catalog
FR-02	Raw data preservation
FR-03	Data cleaning
FR-04	Quality checks
FR-05	Data dictionary
FR-06	Structured storage
FR-07	Reproducible processing
FR-08	Overview dashboard
FR-09	Theme navigation
FR-10	Filtering
FR-11	Time-series charts
FR-12	Theme analysis
FR-13	Metric definitions
FR-14	Source traceability
FR-15	Evidence labels
FR-16	Statistical analysis
FR-17	Forecasting
FR-18	Public opinion
FR-19	Company comparison
FR-20	No-data handling
FR-21	Dashboard guide
FR-22	Version control
FR-23	Issue log
FR-24	Final deliverables
FR-25	Filtered export

1) Non-functional Requirements:

Implement all of these features if possible:

ID	Quality area
NFR-01	Performance
NFR-02	Performance
NFR-03	Accuracy
NFR-04	Integrity
NFR-05	Reliability
NFR-06	Reproducibility
NFR-07	Usability
NFR-08	Readability/accessibility
NFR-09	Security
NFR-10	Privacy/ethics
NFR-11	Licensing
NFR-12	Maintainability
NFR-13	Portability
NFR-14	Compatibility
NFR-15	Scalability
NFR-16	Documentation
NFR-17	Uncertainty transparency
NFR-18	Source resilience
NFR-19	Navigation
NFR-20	Auditability